

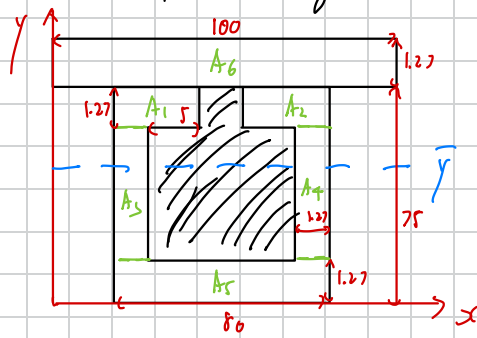
403 Hand Calculations

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First Portion:

Hand Calculations

Find $\bar{y}$, I for design 0:



n) *Note $A_1 = A_2$, $A_3 = A_4$

	①	②	③	④	⑤	⑥
y_{centroid}	74.365mm	74.365mm	37.5mm	37.5mm	0.635mm	75.635mm
$A_{i\text{u}}$	7.9629mm ²	7.9629mm ²	92.0242mm ²	92.0242mm ²	101.6mm ²	127mm ²

b) $\bar{y} = \frac{\sum A_i y_i}{\sum A_i}$

$$= \frac{74.365 \times 7.9629 \times 2 + 37.5 \times 92.0242 \times 2 + 0.635 \times 101.6 + 75.635 \times 127}{7.9629 \times 2 + 92.0242 \times 2 + 101.6 + 127} \quad \frac{\text{mm}^2(\text{mm})}{\text{mm}^2}$$

$$= \frac{17756.3459 \text{ mm}^3}{428.5742 \text{ mm}^2}$$

$$= 41.4512 \dots$$

$$= 41.4 \text{ mm}$$

c.) Apply Parallel axis theorem:

Since all individual areas are rectangles, $I_{\text{rectangle}} = \frac{b h^3}{12}$

	①	②	③	④	⑤	⑥
distance from individual centroid to $\bar{y}$, d_i	32.934mm	32.934mm	3.931mm	3.931mm	41.066mm	34.204mm
$A_{i\text{u}}$	7.9629mm ²	7.9629mm ²	92.0242mm ²	92.0242mm ²	101.6mm ²	127mm ²
b_i	6.27mm	6.27mm	1.27mm	1.27mm	80mm	100mm
h_i	1.27mm	1.27mm	72.46mm	72.46mm	1.27mm	1.27mm
$I_{i\text{u}}$ (mm ⁴)	1.0703	1.0703	40264.0507	40264.0507	13.6559	17.0699

d) $I_{\text{total}} = \sum I_{i\text{u}} + \sum A_i d_i^2$

$$= (1.0703 \times 2 + 40264.0507 \times 2 + 13.6559 + 17.0699) \text{ mm}^4$$

$$+ 7.9629 \times 32.934^2 \times 2 + 92.0242 \times 3.931^2 \times 2 + 101.6 \times 41.066^2 + 127 \times 34.204^2 \text{ mm}^4$$

$$= 80,560.9678 + 33,5614.8518$$

$$= 0.419 \times 10^6 \text{ mm}^4$$

Geometrical properties, Q and b :

For y_{cent} , consider bottom part,

$$Q_{cent} = \sum_{i=1}^n A_i d_i$$

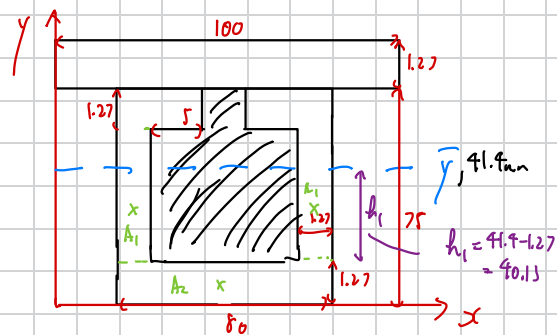
$$= 1.27(40.13)(41.4 - (\frac{40.13}{2} + 1.27)) \times 2 + 80(1.27)(41.4 - (\frac{1.27}{2}))$$

$$= 2041.229463 + 4111.249$$

$$= 6156.973963 \text{ mm}^3$$

$$b = 1.27 \times 2$$

$$= 2.54 \text{ mm}$$



For y_{glue} :

Consider top part,

$$Q = A(d)$$

$$Q_{glue} = 100(1.27)(77 + \frac{1.27}{2} - 41.4)$$

$$= 4347.895 \text{ mm}^3$$

$$b_{glue} = \text{length of glue applied}$$

$$= (7 + 1.27) \times 2$$

$$= 12.54 \text{ mm}$$

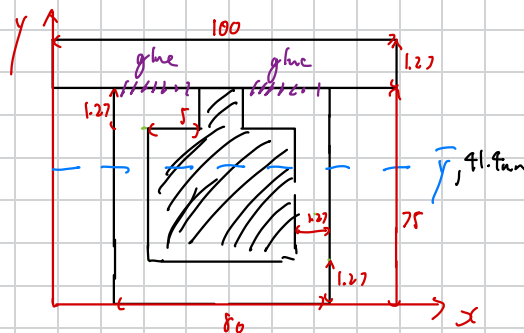
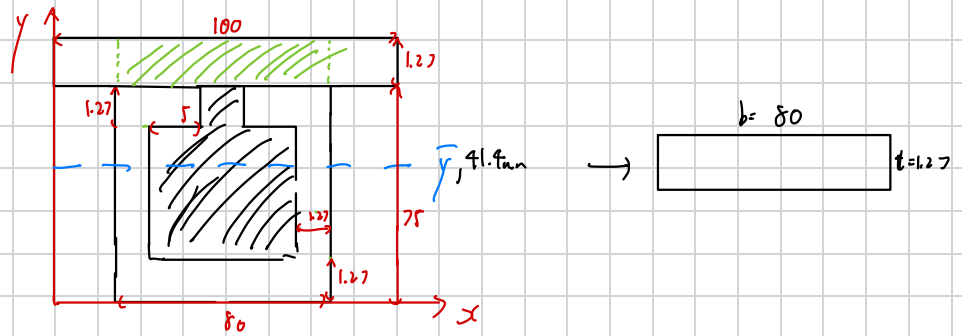


Plate buckling

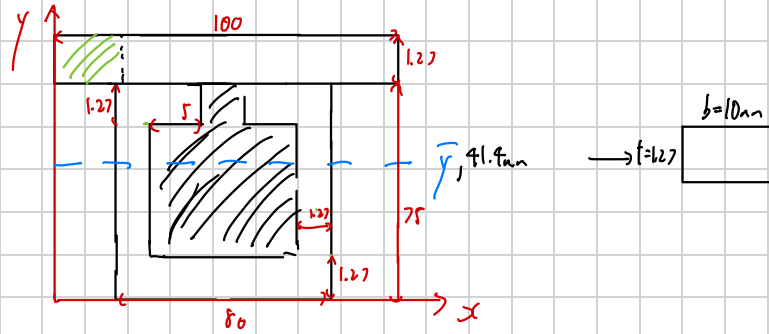
Case 1, top flange w/o side pieces.

$$\begin{aligned}\sigma_{\text{buckling } 1} &= \frac{4\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \\ &= \frac{4\pi^2 (4000)}{12(1-0.2^2)} \left(\frac{1.27}{80}\right)^2 \\ &= 13107.78389 \left(2.520 \times 10^{-9}\right) \\ &= 345 \text{ MPa}\end{aligned}$$



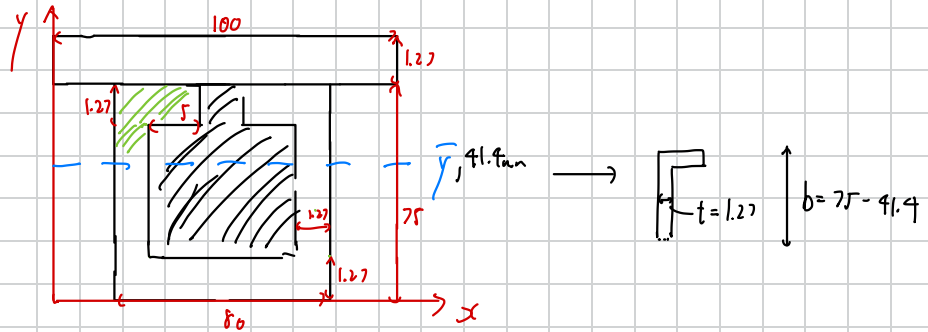
Case 2, free edges

$$\begin{aligned}\sigma_{\text{buckling } 2} &= \frac{0.4254\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \\ &= \frac{0.4254\pi^2 (4000)}{12(1-0.2^2)} \left(\frac{1.27}{10}\right)^2 \\ &= 1457.8228 \left(0.016129\right) \\ &= 23.51 \text{ MPa}\end{aligned}$$



Case 3, side webs

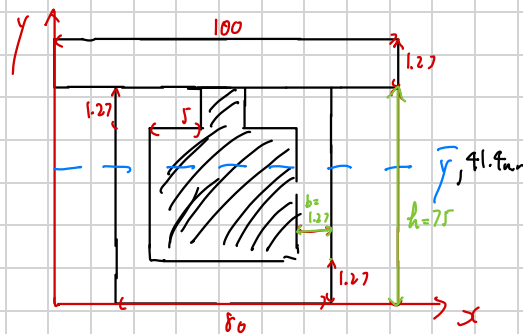
$$\begin{aligned}\sigma_{\text{buckling } 3} &= \frac{6\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \\ &= \frac{6\pi^2 (4000)}{12(1-0.2^2)} \left(\frac{1.27}{75-41.4}\right)^2 \\ &= 20561.6784 \left(1.428660006 \times 10^{-9}\right) \\ &= 29.4 \text{ MPa}\end{aligned}$$



Case 4, shear buckling

Consider the whole cross section,
a = 400mm, spacing of diaphragms

$$\begin{aligned}\tau_{\text{shear buckling}} &= \frac{5\pi^2 E}{12(1-\mu^2)} \left(\left(\frac{b}{a}\right)^2 + \left(\frac{b}{h}\right)^2 \right) \\ &= \frac{5\pi^2 (4000)}{12(1-0.2^2)} \left(\left(\frac{1.27}{75}\right)^2 + \left(\frac{1.27}{400}\right)^2 \right) \\ &= (17134.72986) \left(2.96818 \times 10^{-9}\right) \\ &= 5.09 \text{ MPa}\end{aligned}$$

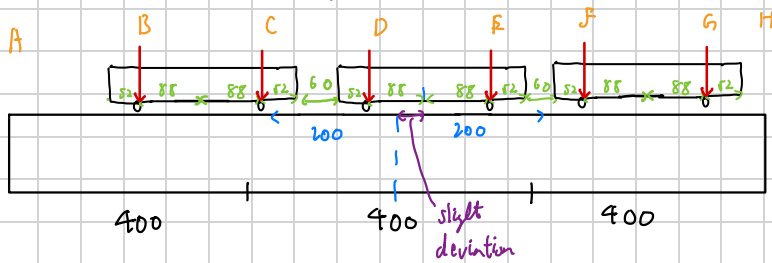


Load Case #1

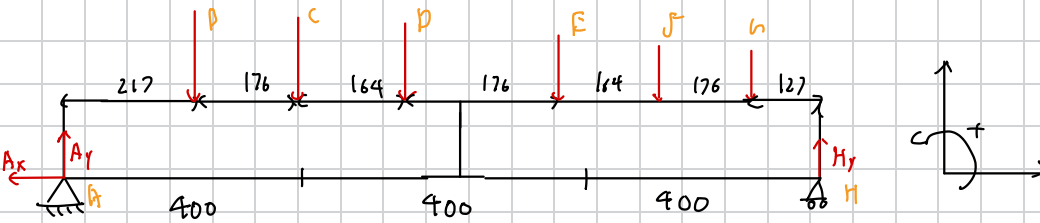
When train is slightly off
the middle
(max bending moment)

①. From the code,
when train is centered at 645mm,
it starts at 212, where max bending moment occurs.

① Dimensions and loading: (not to scale)



② FBD (not to scale)



③ Find reaction forces

$$\sum F_x = 0,$$

$$A_x = 0$$

$$\text{Each load is } \frac{900}{6} = 150 \text{ N}$$

$$\sum M_A = 0$$

$$-\frac{400}{6} (217 + 393 + 557 + 733 + 897 + 1073) + H_y (1200) = 0$$

$$1200 H_y = 258000$$

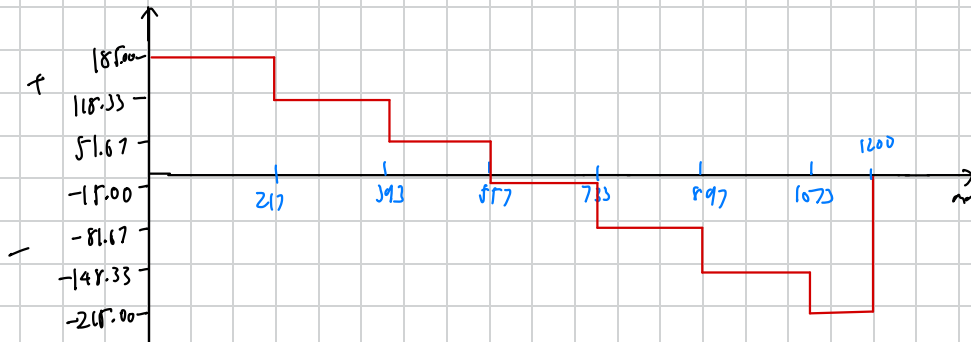
$$H_y = 215 \text{ N}$$

$$\sum F_y = 0$$

$$A_y + H_y = 900$$

$$A_y = 185 \text{ N}$$

④ SFD:



Areas:

$$A_1 = 185 (217) = 40145 \text{ Nmm}$$

$$A_2 = 118.33 (176) = 20826.08 \text{ Nmm}$$

$$A_3 = 51.67 (164) = 8473.88 \text{ Nmm}$$

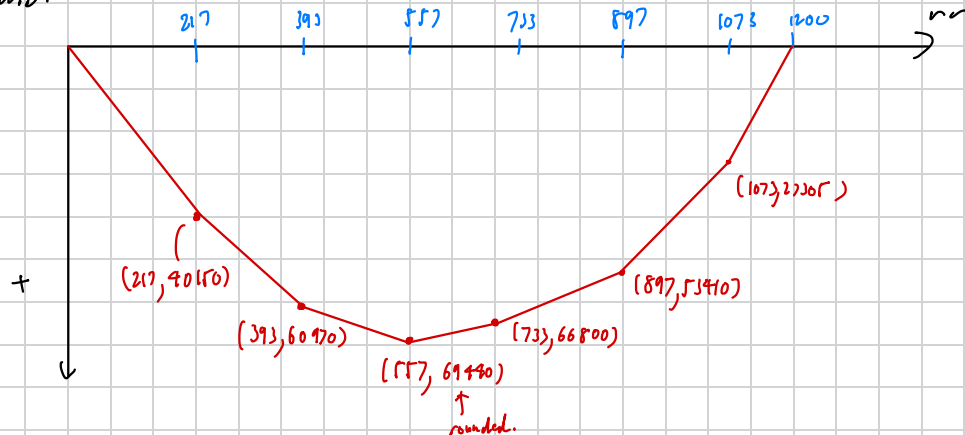
$$A_4 = -15.00 (176) = -2640 \text{ Nmm}$$

$$A_5 = -81.67 (164) = -13393.88 \text{ Nmm}$$

$$A_6 = -148.33 (176) = -26106.08 \text{ Nmm}$$

$$A_7 = -215 (127) = -27305 \text{ Nmm}$$

⑤ BMD:



$$\bar{y} = 41.4 \text{ mm}$$

$$I = 0.419 \times 10^6 \text{ mm}^4$$

⑦ Find bending stresses:

Given top is in compression, bottom is in tension, $I = 0.419 \times 10^6 \text{ mm}^4$

$$y_{\text{top}} = 75 + 1.27 - 41.4$$

$$= 34.8 \text{ mm}$$

$$y_{\text{bot}} = 41.4 \text{ mm}$$

Maximum bending moment occurs @ $x = 857 \text{ mm}$, $M_{\text{max}} = 69440 \text{ Nmm}$

$$\sigma_{\text{top, comp}} = \frac{M y_{\text{top}}}{I} = 5.77 \text{ MPa}$$

$$\sigma_{\text{bot, tension}} = \frac{M y_{\text{bot}}}{I} = 6.86 \text{ MPa}$$

⑧ Find shear stresses:

$$\text{Given } \tau = \frac{VQ}{Ib}$$

$$\text{Take } V_{\text{max}} = 215 \text{ N}, I = 0.419 \times 10^6 \text{ mm}^4$$

⑨ For τ_{cent} , take Q_{cent} and b_{cent}

$$\tau = \frac{215 \text{ N} (618.47)(46.5) \text{ mm}^3}{(0.419 \times 10^6 \text{ mm}^4)(2.59 \text{ mm})}$$

$$\tau_{\text{cent}} = 1.244 \text{ MPa}$$

⑩ For τ_{glue} , take Q_{cent} and b_{cent}

$$= \frac{(215 \text{ N})(4347.845 \text{ mm}^3)}{(0.419 \times 10^6 \text{ mm}^4)(12.59 \text{ mm})}$$

$$= 0.1779 \text{ MPa}$$

⑪ FOS.

$$FOS_{\text{tension}} = \sigma_{\text{tension}} / \sigma_{\text{bot, tension}}$$

$$= 30 / 6.86$$

$$= 4.37 > 1$$

$$FOS_{\text{comp}} = \sigma_{\text{comp}} / \sigma_{\text{top, comp}}$$

$$= 6 / 5.77$$

$$= 1.040 > 1$$

$$FOS_{\text{flexural buckling}} = \min \{ \sigma_{1,2,3 \text{ buckling}} \} / \sigma_{\text{top, comp}}$$

$$= 3.45 / 5.77$$

$$= 0.595 < 1$$

$$FOS_{\text{shear}} = \tau_{\text{max}} / \tau_{\text{cent}}$$

$$= 4 / 1.244$$

$$= 3.22 > 1$$

$$FOS_{\text{glue}} = \tau_{\text{glue, max}} / \tau_{\text{glue}}$$

$$= 2 / 0.1779$$

$$= 11.24 > 1$$

$$FOS_{\text{shear buckling}} = \tau_{\text{shear buckling}} / \tau_{\text{cent}}$$

$$= 5.09 / 1.244$$

$$= 4.09 > 1$$

⑫ Find failing moment for flexural buckling:

$$= FOS (M_{\text{max}})$$

$$= 41500 \text{ Nmm}$$

(Failure due to buckling of flange b/w 2 webs)

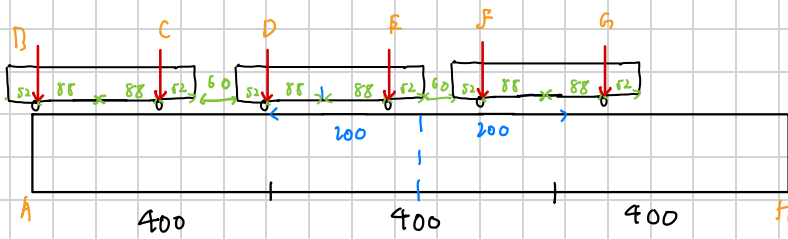
Load

Case

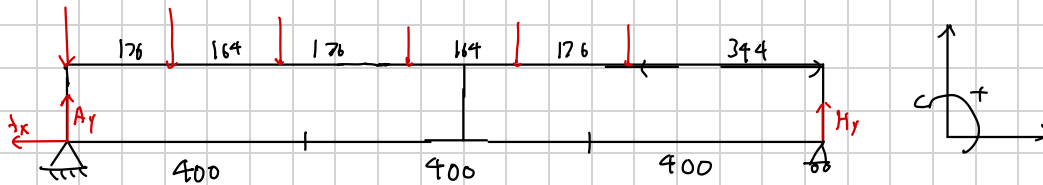
#1

When train is on the edge
(max shear force)

① Dimensions and loading: (Not to scale)



② FBD (Not to scale)



③ Find reaction forces

$$\sum F_x = 0,$$

$$A_x = 0$$

$$\text{each point load} = \frac{400}{6} \text{ N} = 66.67 \text{ N}$$

$$\sum M_A = 0$$

$$-66.67 [176 + 340 + 516 + 680 + 856] + H_y (1200) = 0$$

$$H_y (1200) = 171200$$

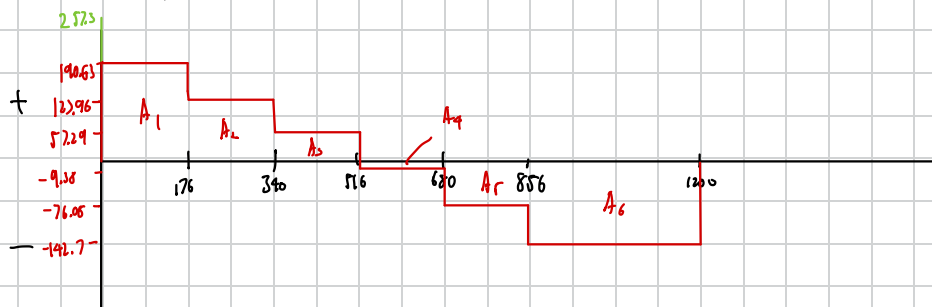
$$H_y = 142.7 \text{ N}$$

$$\sum F_y = 0$$

$$A_y + H_y = 400$$

$$A_y = 257.3 \text{ N}$$

④ SFD:



Areas:

$$A_1 = 33550 \text{ Nmm}$$

$$A_2 = 20330 \text{ Nmm}$$

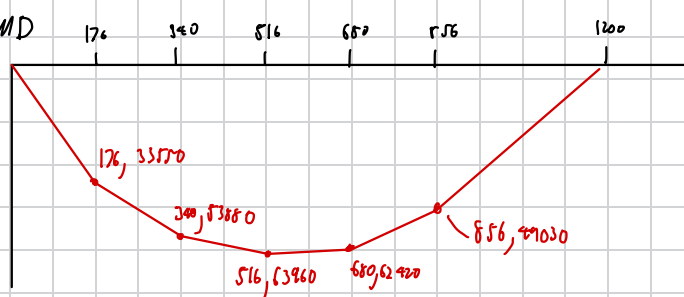
$$A_3 = 10080 \text{ Nmm}$$

$$A_4 = -1540 \text{ Nmm}$$

$$A_5 = -13390 \text{ Nmm}$$

$$A_6 = -49100 \text{ Nmm}$$

⑤ BMD



$$\bar{y} = 41.4 \text{ mm}$$

$$I = 0.419 \times 10^6 \text{ mm}^4$$

⑦ Find bending stresses:

Given top is in compression, bottom is in tension, $I = 0.419 \times 10^6 \text{ mm}^4$

$$y_{\text{top}} = 75 + 1.27 - 41.4$$

$$= 34.8 \text{ mm}$$

$$y_{\text{bot}} = 41.4 \text{ mm}$$

Maximum bending moment occurs @ $x = 516 \text{ mm}$, $M_{\text{max}} = 63960 \text{ Nmm}$

$$\sigma_{\text{top, comp}} = \frac{M y_{\text{top}}}{I} = 5.31 \text{ MPa}$$

$$\sigma_{\text{bot, tension}} = \frac{M y_{\text{bot}}}{I} = 6.32 \text{ MPa}$$

⑧ Find shear stresses:

$$\text{Given } \tau = \frac{VQ}{Ib} \quad (\text{as } x \rightarrow 0)$$

$$\text{Take } V_{\text{max}} = 257.2, I = 0.419 \times 10^6 \text{ mm}^4$$

⑨ For τ_{cent} , take Q_{cent} , b_{cent}

$$\tau = \frac{257.2 (6178.47463) \text{ mm}^3}{0.419 \times 10^6 \text{ mm}^4 (2.74 \text{ mm})}$$

$$\tau_{\text{cent}} = 1.487 \text{ MPa}$$

⑩ For τ_{glue} , take Q_{glue} , b_{glue}

$$\tau_{\text{glue}} = \frac{(257.2) (4397.847 \text{ mm}^3)}{(0.419 \times 10^6 \text{ mm}^4) (12.19 \text{ mm})}$$

$$= 0.213 \text{ MPa}$$

⑪ FOS.

$$FOS_{\text{tension}} = \sigma_{\text{tension}} / \sigma_{\text{bot, tension}}$$

$$= 30 / 6.32$$

$$= 4.75 > 1$$

$$FOS_{\text{flexural buckling}} = \min \{ \sigma_{1,2,3 \text{ buckling}} \} / \sigma_{\text{top, comp}}$$

$$= 3.45 / 5.31$$

$$= 0.650 < 1$$

$$FOS_{\text{glue}} = \tau_{\text{glue, max}} / \tau_{\text{glue}}$$

$$= 2 / 0.213$$

$$= 9.39 > 1$$

$$FOS_{\text{comp}} = \sigma_{\text{comp}} / \sigma_{\text{top, comp}}$$

$$= 6 / 5.31$$

$$= 1.130 > 1$$

$$FOS_{\text{shear}} = \tau_{\text{max}} / \tau_{\text{cent}}$$

$$= 4 / 1.487$$

$$= 2.69 > 1$$

$$FOS_{\text{shear buckling}} = \tau_{\text{shear buckling}} / \tau_{\text{cent}}$$

$$= 0.09 / 1.487$$

$$= 3.42 > 1$$

⑫ Find failing moment for flexural buckling:

$$= FOS (M_{\text{max}})$$

$$= 41600 \text{ Nmm}$$

(Failure due to buckling of flange b/w 2 webs)

In summary, minimum FOS are:

$$FOS_{tension} = 4.37$$

$$FOS_{skewal buckling} = 0.598$$

$$FOS_{glue} = 9.39$$

$$FOS_{comp} = 1.040$$

$$FOS_{shear} = 2.69$$

$$FOS_{shear buckling} = 3.92$$

Second Portion:

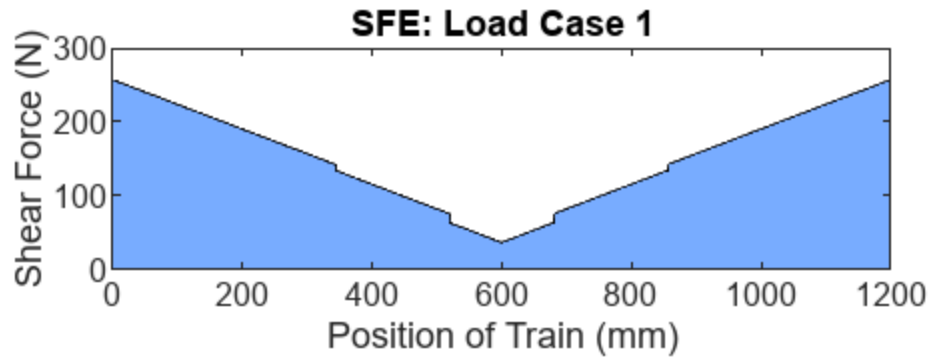
A.) Code output for all intermediate calculations for Design 0 under Load Case 1.

Note the code has outputs for edge and center as it can be applied for calculations for non-constant cross-sectional areas. All "edge" and "center" numbers are equal for constant cross-sectional areas.

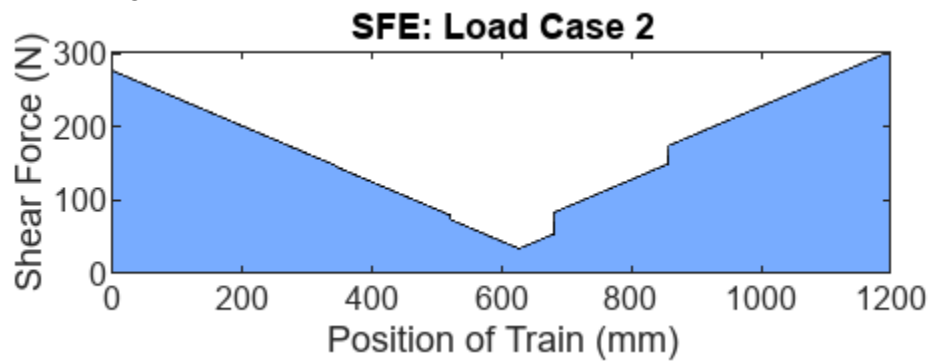
Moment of Inertia for edge: 418352.209
Moment of Inertia for center: 418352.209
y_bar for edge: 41.4311
y_bar for center: 41.4311
Q_cent for edge: 6193.2833
Q_cent for center: 6193.2833
b_cent for edge: 2.54
b_cent for center: 2.54
b_glue for edge: 12.54
b_glue for center: 12.54
Plate buckling stresses for edge: 3.4546, 23.5132, 29.43, 5.0859
Plate buckling stresses for center: 3.4546, 23.5132, 29.43, 5.0859
Maximum Shear Force Load Case 1: 257.3333
Maximum Bending Moment Load Case 1: 69445.3333
Compressive stress Load Case 1: 5.7832
Tensile stress Load Case 1: 6.8774
Max Tau center Load Case 1: 1.4998
Max Tau glue Load Case 1: 0.21308
Maximum Shear Force Load Case 2: 304.1033
Maximum Bending Moment Load Case 2: 78098.9533
Compressive stress Load Case 2: 6.5038
Tensile stress Load Case 2: 7.7345
Max Tau center Load Case 2: 1.7724
Max Tau glue Load Case 2: 0.2518

B.) SFE for Design 0 under Load Case 1, Final Design under Load Case 2.

Design 0, Load Case 1:

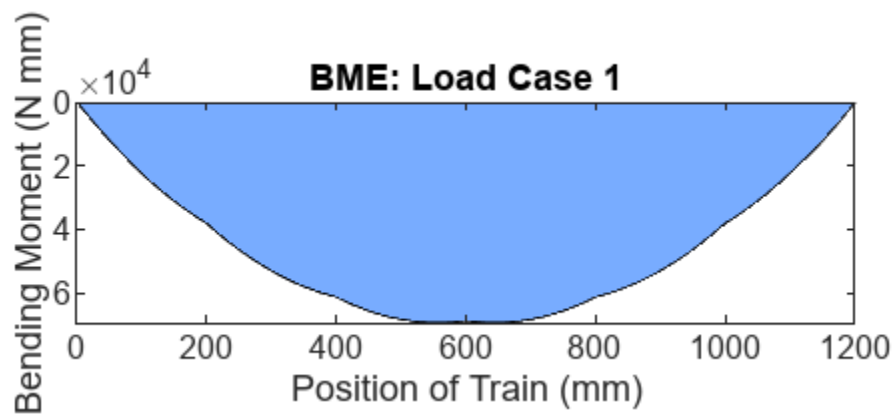


Final Design, Load Case 2:

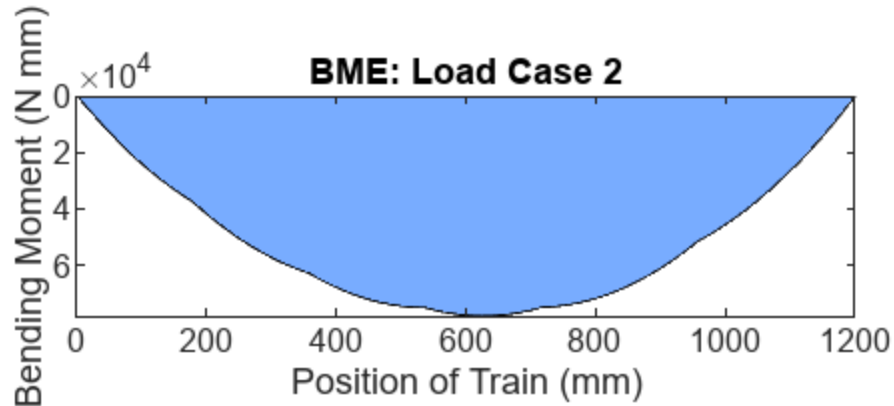


C.) BME for Design 0 under Load Case 1, Final Design under Load Case 2.

Design 0, Load Case 1:



Final Design, Load Case 2:



D.) Code output of FOS for Design 0 under Load Case 1 and Final Design under Load Case 1 and 2.

Design 0, Load Case 1:

Load Case 1
 FOS for tension 4.3621
 FOS for compression: 1.0375
 FOS for flexural buckling: 0.59735
 Case 1 buckling
 FOS for shear: 2.667
 FOS for glue: 9.3863
 FOS for shear buckling: 3.391

Final Design, Load Case 1 and 2:

Load Case 1
 FOS for tension 4.8431
 FOS for compression: 2.3182
 FOS for flexural buckling: 13.6674
 Case 1 buckling
 FOS for shear: 2.7214
 FOS for glue: 11.9087
 FOS for shear buckling: 3.7092
 Load Case 2
 FOS for tension 4.3065
 FOS for compression: 2.0613
 FOS for flexural buckling: 12.153
 Case 1 buckling
 FOS for shear: 2.3029
 FOS for glue: 10.0772
 FOS for shear buckling: 3.1387

E.) Script

Calculator:

Initializing everything

```
clear all %Resets all parameters
```

Initial Parameters

```
n = 1201; % 1201 intervals  
L = 1200; % 1200mm for bridge length  
x = linspace(0, L, n); % Initialize a vector of given intervals  
total_loads = zeros(1, n); % Initialize loading vector
```

Material Properties

```
E = 4000; % Young's modulus  
mu = 0.2; % Poisson's ratio  
ten_strength = 30; % Tensile Strength  
comp_strength = 6; % Compressive Strength  
shear_strength_board = 4; % Shear Strength for Matboard  
shear_strength_glue = 2; % Shear Strength for Glue
```

Point Loading + SFD and MBD

```
x0 = 172; % Left most position of the train (starting position)  
axle_offsets = [0, 176, 340, 516, 680, 856]; % Spacing between  
axles  
axle_positions = x0 + axle_offsets; % Position of axles on bridge
```

```
% Load Case 1:
```

```
P_train = -400; % Total load of the train (N)
```

```
P_train_axle = P_train / 6; % Load per train axle (N)
```



```
%{
[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(1),
P_train_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(2),
P_train_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(3),
P_train_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(4),
P_train_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(5),
P_train_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(6),
P_train_axle, n, total_loads);
%}
```

```
% Load Case 2:

P_locomotive = -182; % Total load of the locomotive (N)
P_freight = -135; % Total load of the freight cars (N)
P_loco_axle = P_locomotive / 2; % Load per locomotive axle (N)
P_freight_axle = P_freight / 2; % Load per freight axle (N)
```

```
% Add point load for each axle

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(1),
P_freight_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(2),
P_freight_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(3),
P_freight_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(4),
P_freight_axle, n, total_loads);

[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(5),
P_loco_axle, n, total_loads);
```

```
[SFD_pl, BMD_pl, total_loads] = point_load(axle_positions(6),  
P_loco_axle, n, total_loads);
```

```
% Plot SFD and BMD  
  
subplot(2, 1, 1);  
  
SFD = area(x, SFD_pl);  
  
SFD.BaseLine.LineStyle = '--';  
  
SFD.FaceColor = "#78acff";  
  
xlabel("x (mm)")  
  
ylabel("Shear Force (N)")  
  
title("SFD")
```

```
  
subplot(2, 1, 2);  
  
BMD = area(x, BMD_pl);  
  
BMD.BaseLine.LineStyle = '--';  
  
BMD.FaceColor = "#78acff";  
  
xlabel("x (mm)")  
  
ylabel("Bending Moment (N mm)")  
  
title("BMD")  
  
set(gca, 'YDir', 'reverse')
```

Load Case 1

```
% Initialize SF and BM envelopes for load case 1  
  
SFE_lc1 = zeros(1, n);  
  
BME_lc1 = zeros(1, n);
```

```
% Train movement loop
```

```

for xi = 1:(L - max(axle_offsets) + 1)

    train_start = xi; % Starting position of the train

    axle_positions = train_start + axle_offsets; % Current axle
positions

    % Initialize SFD and BMD for the current train position

    SFD_current = zeros(1, n);

    BMD_current = zeros(1, n);

    % Loop over axles with different loads
    for i = 1:length(axle_offsets)

        if axle_positions(i) <= L % Axle is on the bridge

            % Reaction forces for point load

            reaction_left = P_train_axle * (L - axle_positions(i)) /
L;

            reaction_right = P_train_axle * axle_positions(i) / L;

            % SF Calculation

            SFD_current = SFD_current + P_train_axle * (x >=
axle_positions(i)) - reaction_left;

            % BM Calculation

            BMD_current = BMD_current + reaction_left .* max(0, x) -
P_train_axle * ...

                max(0, x - axle_positions(i));

        end

    end

    % Update envelopes

```

```
SFE_lc1 = max(SFE_lc1, abs(SFD_current));  
BME_lc1 = max(BME_lc1, abs(BMD_current));  
end
```

```
% Find max shear and moment values and their locations  
shear_max = max(SFE_lc1);  
moment_max = max(BME_lc1);  
x_BM_max_lc1 = find(BME_lc1 == max(BME_lc1));  
x_BM_max_lc1 = x_BM_max_lc1(1);
```

```
figure();  
subplot(2, 1, 1);  
SFE = area(x, SFE_lc1);  
SFE.BaseLine.LineStyle = '--';  
SFE.FaceColor = "#78acff";  
xlabel("Position of Train (mm)")  
ylabel("Shear Force (N)")  
title("SFE: Load Case 1")
```

```
subplot(2, 1, 2);  
BME = area(x, BME_lc1);  
BME.BaseLine.LineStyle = '--';  
BME.FaceColor = "#78acff";  
xlabel("Position of Train (mm)")  
ylabel("Bending Moment (N mm)")  
title("BME: Load Case 1")  
set(gca, 'YDir', 'reverse')
```

Load Case 2

```
% Initialize SF and BM envelopes for load case 2

SFE_lc2 = zeros(1, n);
BME_lc2 = zeros(1, n);

% Train movement loop
for xi = 1:(L - max(axle_offsets) + 1)
    train_start = xi; % Starting position of the train
    axle_positions = train_start + axle_offsets; % Current axle
    positions

    % Initialize SFD and BMD for the current train position
    SFD_current = zeros(1, n);
    BMD_current = zeros(1, n);

    % Loop over axles with different loads
    for i = 1:length(axle_offsets)
        if axle_positions(i) <= L % Axle is on the bridge
            % Assign axle load based on axle index
            if i <= 4 % Freight axles
                axle_load = P_freight_axle;
            else % Locomotive axles
                axle_load = P_loco_axle;
            end

            % Reaction forces for point load
            reaction_left = axle_load * (L - axle_positions(i)) / L;
            reaction_right = axle_load * axle_positions(i) / L;
```

```

        % SF Calculation

        SFD_current = SFD_current + axle_load * (x >=
axle_positions(i)) - reaction_left;

        % BM Calculation

        BMD_current = BMD_current + reaction_left .* max(0, x) -
axle_load * max(0, x - axle_positions(i));

    end

end

% Update envelopes

SFE_lc2 = max(SFE_lc2, abs(SFD_current));

BME_lc2 = max(BME_lc2, abs(BMD_current));

end

```

```

% Find max shear and moment values and their locations

shear_max_lc2 = max(SFE_lc2);

moment_max_lc2 = max(BME_lc2);

x_BM_max_lc2 = find(BME_lc2 == moment_max_lc2);

x_BM_max_lc2 = x_BM_max_lc2(1);

```

```

% Plot results

figure();

subplot(2, 1, 1);

SFE = area(x, SFE_lc2);

SFE.BaseLine.LineStyle = '--';

SFE.FaceColor = "#78acff";

xlabel("Position of Train (mm)")

```

```
ylabel("Shear Force (N)")
title("SFE: Load Case 2")
```

```
subplot(2, 1, 2);
BME = area(x, BME_lc2);
BME.BaseLine.LineStyle = '--';
BME.FaceColor = "#78acff";
xlabel("Position of Train (mm)")
ylabel("Bending Moment (N mm)")
title("BME: Load Case 2")
set(gca, 'YDir', 'reverse')
```

Cross-section Calculations

```
% Edges of bridge (from 0-400 and 800-1200)
x_edge = [0 1200]; % Location, x, of cross-section change
tfw_edge = [100 100]; % Top Flange Width
tfh_edge = [1.27 1.27]; % Top Flange Thickness
wh_edge = [72.46 72.46]; % Web Height
ww_edge = [1.27 1.27]; % Web Thickness (Assuming 2 separate webs)
ws_edge = [80 80]; % Web Spacing
bfh_edge = [1.27 1.27]; % Bottom Flange Height
bfw_edge = [80 80]; % Bottom Flange Width
gtw_edge = [6.27 6.27]; % Glue Tab Width
gth_edge = [1.27 1.27]; % Glue Tab Thickness
a_edge = [400 400]; % Diaphragm Spacing

% Fill in the data for each point on bridge
tfw_edge = interp1(x_edge, tfw_edge, x);
```

```
tfh_edge = interp1(x_edge, tfh_edge, x);
wh_edge = interp1(x_edge, wh_edge, x);
ww_edge = interp1(x_edge, ww_edge, x);
ws_edge = interp1(x_edge, ws_edge, x);
bfh_edge = interp1(x_edge, bfh_edge, x);
bfw_edge = interp1(x_edge, bfw_edge, x);
gtw_edge = interp1(x_edge, gtw_edge, x);
gth_edge = interp1(x_edge, gth_edge, x);
a_edge = interp1(x_edge, a_edge, x);
```

```
% Middle of bridge (from 400-800)
```

```
x_mid = [400 800];
tfw_mid = [100 100];
tfh_mid = [1.27 1.27];
wh_mid = [72.46 72.46];
ww_mid = [1.27 1.27];
ws_mid = [80 80];
bfh_mid = [1.27 1.27];
bfw_mid = [80 80];
gtw_mid = [6.27 6.27];
gth_mid = [1.27 1.27];
a_mid = [400 400];
```

```
tfw_mid = interp1(x_mid, tfw_mid, x);
tfh_mid = interp1(x_mid, tfh_mid, x);
wh_mid = interp1(x_mid, wh_mid, x);
ww_mid = interp1(x_mid, ww_mid, x);
ws_mid = interp1(x_mid, ws_mid, x);
```



```

bfw_mid = interp1(x_mid, bfw_mid, x);
bfh_mid = interp1(x_mid, bfh_mid, x);
gtw_mid = interp1(x_mid, gtw_mid, x);
gth_mid = interp1(x_mid, gth_mid, x);
a_mid = interp1(x_mid, a_mid, x);

```

```

% Combine dimensions of the bridge

top_flange_width = [tfw_edge(1:401) tfw_mid(402:800)
tfw_edge(801:n)];

top_flange_height = [tfh_edge(1:401) tfh_mid(402:800)
tfh_edge(801:n)];

web_height = [wh_edge(1:401) wh_mid(402:800) wh_edge(801:n)];

web_width = [ww_edge(1:401) ww_mid(402:800) ww_edge(801:n)];

web_spacing = [ws_edge(1:401) ws_mid(402:800) ws_edge(801:n)];

bottom_flange_width = [bfw_edge(1:401) bfw_mid(402:800)
bfw_edge(801:n)];

bottom_flange_height = [bfh_edge(1:401) bfh_mid(402:800)
bfh_edge(801:n)];

glue_tab_width = [gtw_edge(1:401) gtw_mid(402:800)
gtw_edge(801:n)];

glue_tab_height = [gth_edge(1:401) gth_mid(402:800)
gth_edge(801:n)];

a = [a_edge(1:401) a_mid(402:800) a_edge(801:n)];

```

```

% Total height of the bridge at every point

height_total = top_flange_height + glue_tab_height + web_height +
bottom_flange_height;

```

```

% Calculate cross-section properties for edges and middle

[y_top_edge, y_bot_edge, I_edge, Q_cent_edge, Q_glue_edge] =
cross_section(x, n, tfw_edge, tfh_edge,...

```

```
wh_edge, ww_edge, bfw_edge, bfh_edge, gtw_edge, gth_edge);
```

```
[y_top_mid, y_bot_mid, I_mid, Q_cent_mid, Q_glue_mid] =  
cross_section(x, n, tfw_mid, tfh_mid, wh_mid, ...  
    ww_mid, bfw_mid, bfh_mid, gtw_mid, gth_mid);
```

```
% Combine properties of the bridge
```

```
y_top = [y_top_edge(1:401) y_top_mid(402:800) y_top_edge(801:n)];
```

```
y_bot = [y_bot_edge(1:401) y_bot_mid(402:800) y_bot_edge(801:n)];
```

```
I = [I_edge(1:401) I_mid(402:800) I_edge(801:n)];
```

```
Q_cent = [Q_cent_edge(1:401) Q_cent_mid(402:800)  
Q_cent_edge(801:n)];
```

```
Q_glue = [Q_glue_edge(1:401) Q_glue_mid(402:800)  
Q_glue_edge(801:n)];
```

Thin Plate Buckling (Same for every case)

```
% Case 1 - restrained on both sides
```

```
c1_height = top_flange_height;
```

```
c1_width = web_spacing;
```

```
case1_stress = thin_plate_buckling1(x, n, E, mu, c1_height,  
c1_width);
```

```
% Case 2 - restrained on one side
```

```
c2_height = top_flange_height;
```

```
c2_width = (top_flange_width - web_spacing) / 2;
```

```
case2_stress = thin_plate_buckling2(x, n, E, mu, c2_height,  
c2_width);
```

```
% Case 3 - variable stress
```

```
c3_height = web_width;
```

```

c3_width = (bottom_flange_height + web_height + glue_tab_height) -
y_bot;

case3_stress = thin_plate_buckling3(x, n, E, mu, c3_height,
c3_width);

```

```

% Case 4 - shear

```

```

t = web_width;

h = web_height + glue_tab_height + bottom_flange_height;

shear_stress = shear_buckling(x, n, E, mu, t, h, a);

```

Load Case 1 FOS

```

comp_stress_lc1 = max(BME_lc1) * y_top(x_BM_max_lc1) /
I(x_BM_max_lc1); % Stress at top

ten_stress_lc1 = max(BME_lc1) * y_bot(x_BM_max_lc1) /
I(x_BM_max_lc1); %Stress at bottom

```

```

V_lc1 = SFE_lc1;

b_cent = web_width * 2;

b_glue = glue_tab_width * 2;

```

```

tau_cent_lc1 = tau(x, n, V_lc1, Q_cent, I, b_cent);

tau_glue_lc1 = tau(x, n, V_lc1, Q_glue, I, b_glue);

```

```

% Calculate FOS

FOS_ten_lc1 = ten_strength / ten_stress_lc1;

FOS_comp_lc1 = comp_strength / comp_stress_lc1;

FOS_flexural_buckling_lc1 =
min([min(case1_stress),min(case2_stress),min(case3_stress)]) /
comp_stress_lc1;

FOS_shear_lc1 = shear_strength_board / max(tau_cent_lc1);

```

```
FOS_glue_lc1 = shear_strength_glue / max(tau_glue_lc1);
FOS_shear_buckling_lc1 = shear_stress / max(tau_cent_lc1);
```

```
% Display FOS
disp("Load Case 1")
disp("FOS for tension " + FOS_ten_lc1)
disp("FOS for compression: " + FOS_comp_lc1 )
disp("FOS for flexural buckling: " + FOS_flexural_buckling_lc1)
if min([min(case1_stress),min(case2_stress),min(case3_stress)]) ==
min(case1_stress)
    disp("Case 1 buckling")
elseif min([min(case1_stress),min(case2_stress),min(case3_stress)])
== min(case2_stress)
    disp("Case 2 buckling")
else
    disp("Case 3 buckling")
end
disp("FOS for shear: " + FOS_shear_lc1)
disp("FOS for glue: " + FOS_glue_lc1)
disp("FOS for shear buckling: " + min(FOS_shear_buckling_lc1))
```

Load Case 2 FOS

```
comp_stress_lc2 = max(BME_lc2) * y_top(x_BM_max_lc2) /
I(x_BM_max_lc2); % Stress at top
ten_stress_lc2 = max(BME_lc2) * y_bot(x_BM_max_lc2) /
I(x_BM_max_lc2); %Stress at bottom
```

```
V_lc2 = SFE_lc2;
b_cent = web_width * 2;
```

```
b_glue = glue_tab_width * 2;
```

```
tau_cent_lc2 = tau(x, n, V_lc2, Q_cent, I, b_cent);
```

```
tau_glue_lc2 = tau(x, n, V_lc2, Q_glue, I, b_glue);
```

```
% Calculate FOS
```

```
FOS_ten_lc2 = ten_strength / ten_stress_lc2;
```

```
FOS_comp_lc2 = comp_strength / comp_stress_lc2;
```

```
FOS_flexural_buckling_lc2 =  
min([min(case1_stress),min(case2_stress),min(case3_stress)]) /  
comp_stress_lc2;
```

```
FOS_shear_lc2 = shear_strength_board / max(tau_cent_lc2);
```

```
FOS_glue_lc2 = shear_strength_glue / max(tau_glue_lc2);
```

```
FOS_shear_buckling_lc2 = shear_stress / max(tau_cent_lc2);
```

```
% Display FOS
```

```
disp(" ")
```

```
disp("Load Case 2")
```

```
disp("FOS for tension " + FOS_ten_lc2)
```

```
disp("FOS for compression: " + FOS_comp_lc2 )
```

```
disp("FOS for flexural buckling: " + FOS_flexural_buckling_lc2)
```

```
if min([min(case1_stress),min(case2_stress),min(case3_stress)]) ==  
min(case1_stress)
```

```
    disp("Case 1 buckling")
```

```
elseif min([min(case1_stress),min(case2_stress),min(case3_stress)])  
== min(case2_stress)
```

```
    disp("Case 2 buckling")
```

```
else
```

```
    disp("Case 3 buckling")
```

```

end

disp("FOS for shear: " + FOS_shear_lc2)

disp("FOS for glue: " + FOS_glue_lc2)

disp("FOS for shear buckling: " + min(FOS_shear_buckling_lc2))

```

Displaying Intermediate Calculations

```

disp("Moment of Inertia for edge: " + I(200))

disp("Moment of Inertia for center: " + I(600))

disp("y_bar for edge: " + y_bot(200))

disp("y_bar for center: " + y_bot(600))

disp("Q_cent for edge: " + Q_cent(200))

disp("Q_cent for center: " + Q_cent(600))

disp("b_cent for edge: " + b_cent(200))

disp("b_cent for center: " + b_cent(600))

disp("b_glue for edge: " + b_glue(200))

disp("b_glue for center: " + b_glue(600))

disp("Plate buckling stresses for edge: " + case1_stress(200) + ",
" + case2_stress(200) + ", " + case3_stress(200) + ", " +
shear_stress(200))

disp("Plate buckling stresses for center: " + case1_stress(600) +
", " + case2_stress(600) + ", " + case3_stress(600) + ", " +
shear_stress(600))

disp("Maximum Shear Force Load Case 1: " + max(SFE_lc1))

disp("Maximum Bending Moment Load Case 1: " + max(BME_lc1))

disp("Compressive stress Load Case 1: " + comp_stress_lc1)

disp("Tensile stress Load Case 1: " + ten_stress_lc1)

disp("Max Tau center Load Case 1: " + max(tau_cent_lc1) )

disp("Max Tau glue Load Case 1: " + max(tau_glue_lc1) )

disp("Maximum Shear Force Load Case 2: " + max(SFE_lc2))

```

```

disp("Maximum Bending Moment Load Case 2: " + max(BME_lc2))
disp("Compressive stress Load Case 2: " + comp_stress_lc2)
disp("Tensile stress Load Case 2: " + ten_stress_lc2)
disp("Max Tau center Load Case 2: " + max(tau_cent_lc2))
disp("Max Tau glue Load Case 2: " + max(tau_glue_lc2))

```

Functions

```

% Applying point load
function [SFD_pl BMD_pl total_loads] = point_load(xP, P, n, total_loads)

    xP_B = 1200;

    temp_P_B = -P * xP / xP_B;

    P_A = total_loads(1) - P - temp_P_B;

    P_B = temp_P_B + total_loads(xP_B+1);

```

```

total_loads(1) = P_A;

total_loads(xP+1) = P;

```

```

if xP == xP_B

    total_loads(xP_B+1) = P_B + P;

else

    total_loads(xP_B+1) = P_B;

end

```

```

SFD_pl = zeros(1, n);

SFD_pl(1) = total_loads(1);

```

```

for i = 2:length(total_loads)

    SFD_pl(i) = total_loads(i) + SFD_pl(i-1);

```

```
end
```

```
BMD_pl = cumsum(SFD_pl);
```

```
BMD_pl(n) = 0;
```

```
end
```

```
% Calculating cross-section properties
```

```
function [y_top, y_bot, I, Q_cent, Q_glue] = cross_section(x, n, tfw,  
tfh, ...
```

```
wh, ww, bfw, bfh, gtw, gth)
```

```
% tfw - top flange width
```

```
% tfh - top flange height
```

```
% wh - web height
```

```
% ww - web width
```

```
% bfw - bottom flange width
```

```
% bfh - bottom flange height
```

```
% gtw - glue tab width
```

```
% gth - glue tab height
```

```
total_height = tfh + wh + bfh + gth;
```

```
tf_area = zeros(1, n);
```

```
bf_area = zeros(1, n);
```

```
w_area = zeros(1, n);
```

```
gt_area = zeros(1, n);
```

```
tf_I = zeros(1, n);
```

```
bf_I = zeros(1, n);
```



```
w_I = zeros(1, n);  
gt_I = zeros(1, n);
```

```
tf_Isum = zeros(1, n);  
bf_Isum = zeros(1, n);  
w_Isum = zeros(1, n);  
gt_Isum = zeros(1, n);
```

```
y_top = zeros(1, n);  
y_bot = zeros(1, n);  
I = zeros(1, n);
```

```
w_Q = zeros(1, n);  
Q_cent = zeros(1, n);  
Q_glue = zeros(1, n);
```

```
for i = 1:length(x)  
    tf_area(i) = tfw(i) * tfh(i);  
    bf_area(i) = bfw(i) * bfh(i);  
    w_area(i) = 2 * wh(i) * ww(i);  
    gt_area(i) = 2 * gtw(i) * gth(i);
```

```
    tf_y = total_height - (tfh ./ 2);  
    bf_y = bfh ./ 2;  
    w_y = (wh ./ 2) + bfh;  
    gt_y = total_height(i) - tfh(i) - (gth ./ 2);
```

```
    y_bot(i) = (tf_area(i) * tf_y(i) + bf_area(i) * bf_y(i) +  
w_area(i) * w_y(i) + ...
```

```

        gt_area(i) * gt_y(i)) / (tf_area(i) + bf_area(i) + w_area(i)
+ gt_area(i));

```

```

        y_top(i) = abs(y_bot(i) - total_height(i));

```

```

        tf_I(i) = (tfw(i) * (tfh(i)^3)) / 12;

```

```

        bf_I(i) = (bfw(i) * (bfh(i)^3)) / 12;

```

```

        w_I(i) = 2 * (ww(i) * (wh(i)^3)) / 12;

```

```

        gt_I(i) = 2 * (gtw(i) * (gth(i)^3)) / 12;

```

```

        tf_Isum(i) = tf_I(i) + tf_area(i) * (abs(y_bot(i) - tf_y(i)))^2;

```

```

        bf_Isum(i) = bf_I(i) + bf_area(i) * (abs(y_bot(i) - bf_y(i)))^2;

```

```

        w_Isum(i) = w_I(i) + w_area(i) * (abs(y_bot(i) - w_y(i)))^2;

```

```

        gt_Isum(i) = gt_I(i) + gt_area(i) * (abs(y_bot(i) - gt_y(i)))^2;

```

```

        I(i) = tf_Isum(i) + bf_Isum(i) + w_Isum(i) + gt_Isum(i);

```

```

        w_Q(i) = 2 * (ww(i) * (y_bot(i) - bfh(i))) * (y_bot(i) -
((y_bot(i) - bfh(i)) / 2) + bfh(i));

```

```

        Q_cent(i) = bf_area(i) * (y_bot(i) - (bfh(i)/2)) + w_Q(i);

```

```

        Q_glue(i) = tf_area(i) * (tf_y(i) - y_bot(i));

```

```

    end

```

```

end

```

```

% Tau calulation

```

```

function t = tau(x, n, V, Q, I, b)

```

```

    t = zeros(1, n);

```

```

    for i = 1:length(x)

```

```

        t(i) = (V(i) * Q(i)) / (I(i) * b(i));

```

```
end  
  
end
```

```
% Thin plate buckling  
function tpb1_stress = thin_plate_buckling1(x, n, E, mu, t, b)  
    tpb1_stress = zeros(1, n);
```

```
    for i = 1:length(x)  
        tpb1_stress(i) = ((4 * 3.14159^2 * E) / (12 * (1 - mu^2))) *  
            ((t(i) / b(i)) ^ 2);  
    end  
end
```

```
function tpb2_stress = thin_plate_buckling2(x, n, E, mu, t, b)  
    tpb2_stress = zeros(1, n);
```

```
    for i = 1:length(x)  
        tpb2_stress(i) = ((0.4254 * 3.14159^2 * E) / (12 * (1 - mu^2))) *  
            ((t(i) / b(i)) ^ 2);  
    end  
end
```

```
function tpb3_stress = thin_plate_buckling3(x, n, E, mu, t, b)  
    tpb3_stress = zeros(1, n);
```

```
    for i = 1:length(x)  
        tpb3_stress(i) = ((6 * 3.14159^2 * E) / (12 * (1 - mu^2))) *  
            ((t(i) / b(i)) ^ 2);  
    end
```

```
end
```

```
% Shear buckling
```

```
function ss = shear_buckling(x, n, E, mu, t, h, a)
```

```
    ss = zeros(1, n);
```

```
    for i = 1:length(x)
```

```
        ss(i) = ((5 * (3.14159^2) * E) / (12 * (1 - mu^2))) * (((t(i) /  
h(i)) ^ 2) + ((t(i) / a(i)) ^ 2));
```

```
    end
```

```
end
```

Dataset

Design0

```
% Edges of bridge (from 0-400 and 800-1200)
```

```
x_edge = [0 1200]; % Location, x, of cross-section change
```

```
tfw_edge = [100 100]; % Top Flange Width
```

```
tfh_edge = [1.27 1.27]; % Top Flange Thickness
```

```
wh_edge = [72.46 72.46]; % Web Height
```

```
ww_edge = [1.27 1.27]; % Web Thickness (Assuming 2 separate webs)
```

```
ws_edge = [80 80]; % Web Spacing
```

```
bfbh_edge = [1.27 1.27]; % Bottom Flange Height
```

```
bfbw_edge = [80 80]; % Bottom Flange Width
```

```
gtw_edge = [6.27 6.27]; % Glue Tab Width
```

```
gth_edge = [1.27 1.27]; % Glue Tab Thickness
```

```
a_edge = [400 400]; % Diaphragm Spacing
```

```
% Middle of bridge (from 400-800)
```

```
x_mid = [400 800];
```

```
tfw_mid = [100 100];
```

```
tfh_mid = [1.27 1.27];
```

```
wh_mid = [72.46 72.46];
```

```
ww_mid = [1.27 1.27];
```

```
ws_mid = [80 80];
```

```
bfh_mid = [1.27 1.27];
```

```
bfw_mid = [80 80];
```

```
gtw_mid = [6.27 6.27];
```

```
gth_mid = [1.27 1.27];
```

```
a_mid = [400 400];
```

Design5

```
% Edges of bridge (from 0-400 and 800-1200)
```

```
x_edge = [0 1200]; % Location, x, of cross-section change
```

```
tfw_edge = [100 100]; % Top Flange Width
```

```
tfh_edge = [3.81 3.81]; % Top Flange Thickness
```

```
wh_edge = [73.65 73.65]; % Web Height
```

```
ww_edge = [1.27 1.27]; % Web Thickness (Assuming 2 separate webs)
```

```
ws_edge = [75 75]; % Web Spacing
```

```
bfh_edge = [1.27 1.27 ]; % Bottom Flange Height
```

```
bfw_edge = [75 75]; % Bottom Flange Width
```

```
gtw_edge = [10 10]; % Glue Tab Width
```

```
gth_edge = [1.27 1.27]; % Glue Tab Thickness
```

```
a_edge = [200 200]; % Diaphragm Spacing
```

```
% Middle of bridge (from 400-800)
```

```
x_mid = [400 800];
```

```
tfw_mid = [100 100];  
tfh_mid = [3.81 3.81];  
wh_mid = [73.65 73.65];  
ww_mid = [1.27 1.27];  
ws_mid = [75 75];  
bfh_mid = [1.27 1.27];  
bfw_mid = [75 75];  
gtw_mid = [10 10];  
gth_mid = [1.27 1.27];  
a_mid = [200 200];
```